How HUD Sizes a *223(f) Multifamily Mortgage*

A Former Senior MAP Underwriter Explains the Methodology

When borrowers ask us how large a HUD 223(f) loan can get, the honest answer is: it depends on which of four independent tests is binding. HUD doesn't size a mortgage the way a conventional lender does — it runs multiple constraints simultaneously, and the loan is the lowest result across all of them. This article walks through each test, explains the math behind it, and covers why the binding constraint shifts as interest rates move. There's also a sizing tool at the bottom where you can play around with the math.

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Centennial Mortgage, Inc.
Published April 2026
Last updated October 2026

Current as of October 2026.
Latest version: roachlorenz.com/resources/how-hud-sizes-a-223f-mortgage/

I spent years underwriting HUD loans at a MAP lender before moving into originations, and I've built the Excel underwriting models for nearly every lender I've worked for — including one of the top 20 largest banks in the country — which means I've run these calculations several hundred times from the inside. One of the most persistent misconceptions I encounter is that borrowers — and sometimes brokers — treat HUD loan sizing as a single-variable problem. They'll say "the property is worth $20 million, so at 80% we should get $16 million." That may or may not be right. The 80% LTV figure is one ceiling. There are three others that can cut below it, and in a high-rate environment the one that actually binds often has nothing to do with value.

01

PROGRAM PARAMETERS

The 223(f) at a Glance — *and How It Compares*

The HUD 223(f) program provides FHA mortgage insurance for the acquisition or refinance of existing multifamily properties — no construction involved. It's a 35-year fully amortizing loan at a fixed rate, non-recourse to the sponsors.

PARAMETER	HUD 223(F)	FNMA / FREDDIE (TYPICAL)
Max LTV — Standard	**87%**	75%
Max LTV — Cash-Out	**80%**	70–75%
Min DSCR	**1.15x**	1.25x
Loan Term	**35 years**	5–12 years
Amortization	**35 years**	30 years (up to 35)
Recourse	**Non-recourse**	Non-recourse (with carveouts)
Rate Structure	**Fixed**	Fixed or floating

Fannie Mae & Freddie Mac terms are typical of high leverage transactions. Some strong borrowers and/or markets may be different.

02

A CRITICAL DISTINCTION

Market NOI vs. *Debt Service NOI*

Before walking through the four sizing tests, it's worth explaining a distinction that catches borrowers off guard: HUD uses two separate NOI figures, and they feed different parts of the analysis.

Market NOI is what the property stabilizes to at market occupancy with normalized operating expenses. This is the NOI the third-party appraiser develops alongside the appraiser's sales comp valuation, and it drives the LTV calculation. That is, Market NOI divided by the cap rate equals the appraised value, and the LTV tests apply to that value.

Debt Service NOI is generally a more conservative figure that HUD derives from analyzing the property's actual operating history and the appraiser's NOI to determine what the project's actual expected NOI will be. In addition, HUD requires that the vacancy be 7% for market-rate deals (can be lower if affordable), even if the actual vacancy is lower. It also applies the actual annual replacement reserve escrow amount as determined by the Property Capital Needs Assessment (as opposed to the $250/unit escrow the appraiser generally puts in). While the Market NOI feeds the LTV loan amounts, this Debt Service NOI feeds into the DSCR test directly.

The sizing tool later in this article has separate input fields for Market NOI and Debt Service NOI. If you're working from preliminary estimates and haven't separated these yet, use the same figure in both fields to start — but understand that HUD's Debt Service NOI will almost always be lower than your underwritten market NOI

once third-party reports are in. We break down exactly where the two diverge — across income, vacancy, and expenses — and what that gap is worth, in our companion article on DSCR NOI vs. appraised-value NOI.

03

CONSTRAINT ONE

The LTV Test — *Value-Driven Ceiling*

The LTV test is the most intuitive of the four constraints. HUD commissions an independent appraisal, which establishes market value. The appraiser applies a market cap rate to the stabilized Market NOI to help derive value, and the 87% LTV ceiling is applied to that appraised value.

ILLUSTRATIVE EXAMPLE — LTV TEST

Market NOI: $1,200,000 | **Cap Rate:** 5.50%

`Value = $1,200,000 / 0.055 =` **`$21,818,182`**

`LTV Maximum (87%) = $21,818,182 × 0.87 =` **`$18,981,818`**

The LTV test produces a maximum mortgage of approximately **$18.98 million**. We'll carry this figure into the DSCR example below.

Because Market NOI drives value, cap rate assumption has a significant effect on the LTV-constrained loan amount. A 25-basis-point difference in cap rate on a $1.2 million NOI deal moves value by roughly $1 million — and the LTV ceiling translates that directly into proceeds. This is one reason HUD's third-party appraisal matters so much and why we spend time discussing cap rate selection with borrowers before their appraisal engagement.

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CONSTRAINT TWO

The DSCR Test — *Where Rate Sensitivity Lives*

The DSCR test is where interest rate assumptions have their entire effect on sizing. The mechanics are a bit different from how most lenders describe them, and there's a quirk worth understanding.

The way HUD actually sizes the DSCR-constrained loan is to multiply the Debt Service NOI by 87%. That result is the maximum allowable annual debt service — covering P&I, MIP, and all. The industry shorthand is "1.15x DSCR," which is how most originators and borrowers describe the requirement. But 1 / 0.87 equals approximately 1.149, which rounds to 1.15 — so the 1.15x figure is a convenient approximation of the same underlying calculation.

To convert that debt service budget into a loan amount, you divide by the annual loan constant. The loan constant is the sum of three components: the interest rate, the annual MIP rate, and the amortization rate. The amortization rate is the annual principal repayment per dollar of loan balance at a given rate and term — it's what's left of the annual P&I payment after stripping out the interest component. At 5.00% on a 35-year term, the amortization rate is approximately 1.056%. Add the 0.25% MIP and the 5.00% interest rate and the full loan constant is approximately 6.306%. Divide the DS NOI × 0.87 figure by that constant and you have the DSCR-constrained loan amount.

ILLUSTRATIVE EXAMPLE — DSCR TEST

Debt Service NOI: $1,080,000 | **Rate:** 5.00% | **Amortization:** 35 years | **MIP:** 0.25%/yr

Max Annual Debt Service (P&I + MIP) = $1,080,000 × 0.87 = **$939,600**

Amortization rate (at 5.00%, 35yr) = annual P&I per $1 − 5.00% = **1.056%**

Loan Constant = 5.00% (interest) + 0.25% (MIP) + 1.056% (amortization) = **6.306%**

DSCR Loan Amount = $939,600 / 0.06306 = **$14,900,000**

The DSCR test at 5.00% produces a maximum mortgage of approximately **$14.9 million** — well below the **$18.98 million** LTV ceiling from the prior example. LTV is not the binding constraint here.

At 3.00%, the amortization rate drops to roughly 0.762%, producing a loan constant of approximately 4.012%. The same $939,600 debt service budget supports a loan of roughly **$19.3 million** — which exceeds the LTV ceiling and makes LTV the binding constraint instead. The crossover point where DSCR and LTV produce the same loan is around 3.12% at these assumptions. Below that rate, further rate decreases stop moving proceeds.

The 35-year amortization term is part of why HUD's DSCR test can produce more proceeds than agency lenders at equivalent rates. A 30-year amortization on the same loan amount produces a higher monthly P&I, which consumes more of the debt service budget. The longer HUD amortization schedules a lower monthly constant, stretching a given NOI to support a larger balance. Combined with the 1.15x minimum versus the 1.25x Fannie/Freddie minimum, the cumulative effect on proceeds can be substantial — particularly on lower-NOI deals.

Rate environment and the binding constraint: In a low-rate environment, the DSCR test frequently produces a higher loan amount than the LTV ceiling, making LTV the binding constraint and making rate changes below a threshold mostly irrelevant to proceeds. In a higher-rate environment, DSCR bites first. If you're sizing a deal at 6.00% and re-running it at 5.00%, the proceeds difference often runs $1–2 million on a mid-market property — the entire difference lives in the DSCR test.

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CONSTRAINT THREE

The HUD Statutory Mortgage Limit — *Per-Unit Dollar Caps*

The third constraint is the one borrowers are least familiar with: HUD publishes statutory per-unit mortgage limits by bedroom count that establish an absolute dollar ceiling on the loan, independent of both value and income. These limits exist because HUD mortgage insurance is a federal program with congressional authorization tied to specific loan ceiling amounts.

The base limits are set by statute and adjusted upward via a High Cost Percentage (HCP) factor — currently 270% uniformly across all markets. HUD adds 100% of the appraised land value to the statutory ceiling and 100% of the cost of building space that is not attributable to residential use — referred to as "Cost Not Attributable." This includes commercial square footage, standalone community buildings, hallways, and other non-residential components.

ILLUSTRATIVE EXAMPLE — STATUTORY TEST (ELEVATOR BUILDING)

Using elevator per-unit base limits (approximate; confirm current figures with HUD) and the same property from the LTV example (appraised value $21,818,182):

1-BR: ~$88,832 | 2-BR: ~$108,925 | 3-BR: ~$136,424 | 4+BR: ~$154,259 | 0-BR: ~$80,170

A 100-unit elevator property with 60 one-bedroom and 40 two-bedroom units would calculate as:

(60 × $88,832 × 2.70) + (40 × $108,925 × 2.70) = $14,390,784 + $11,763,900 = **$26,154,684**

Adding Cost Not Attributable (10% of appraised value) and 100% of land value:

Cost Not Attributable = $21,818,182 × 10% = **$2,181,818**

Statutory Maximum = $26,154,684 + $2,181,818 + $2,000,000 (land) = **$30,336,502**

In practice, the statutory limit rarely constrains a 223(f). The deals where it bites are not large-unit properties in average markets — they are properties in very high-rent metros like Manhattan or San Francisco, where both the LTV and DSCR tests would otherwise produce a very large loan. In those markets the income and value support an outsized mortgage, and the statutory per-unit cap is what ultimately cuts the number. The reason this happens infrequently even in high-cost markets is that land values in those same metros tend to be substantial, and adding 100% of land value to the statutory ceiling pushes the limit high enough that it doesn't bind. It is worth running the test, but it should not drive early-stage deal decisions in most markets.

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CONSTRAINT FOUR

The Cost-of-Refinance Criterion — *How HUD Handles Cash-Out*

The fourth criterion works differently from the first three, and it's frequently misunderstood. It isn't simply a floor that the loan has to clear — it's the mechanism HUD uses to govern whether a refinance produces cash-out, and if it does, it caps the LTV at 80%.

Here's how it works: this criterion sizes the loan at the greater of 80% LTV or the total cost of the refinance transaction. The cost of refinance includes the existing debt payoff, any prepayment penalty, required repairs, replacement reserve escrow deposits, third-party report costs, lender fees, HUD fees, and legal costs.

If the cost of refinancing is less than 80% LTV — meaning the property has enough equity that even at 80% the proceeds exceed what it costs to close — the loan produces cash-out. The borrower walks away from closing with funds in excess of the transaction costs. This is the normal refinance outcome on a well-valued, low-debt property.

If the cost of refinancing exceeds 80% LTV, HUD will size the loan to exactly cover those costs — but only up to the 87% LTV ceiling from Criterion One. A refinance where the gross loan lands between the cost of refinance and 87% LTV produces no cash-out but also doesn't require the borrower to bring cash to closing. A refinance where even 87% LTV isn't enough to cover the cost of refinancing is called a cash-in refinance — the borrower has to contribute funds at closing to make the deal work, which is uncommon but possible on heavily leveraged assets with significant repair needs.

The practical interaction between Criterion One (87% LTV) and this criterion is that **HUD has a built-in 80% LTV cash-out cap**. If a deal is producing cash-out, the loan cannot exceed 80% of appraised value — the moment the cost of refinance is fully covered and there's money left over, the applicable LTV ceiling drops from 87% to 80%. Borrowers sometimes ask why HUD caps cash-out at a lower LTV than a standard refinance — this is why. The 87% ceiling is reserved for deals where the proceeds are consumed by the transaction itself.

The sizing tool below includes fields for existing debt payoff, prepayment penalty, and estimated repair escrow so you can model where this criterion lands relative to the upper constraints.

07

PUTTING IT TOGETHER

Which Test Bites — *and When*

The actual loan amount is the lowest result across all four tests. In practice, the pattern runs roughly like this:

- **High-rate environments (roughly 5.50%+):** The DSCR test typically produces the lowest number. The same NOI budget supports a smaller loan at higher rates, so DSCR bites before LTV. Improving the DSCR test means increasing Debt Service NOI — rent increases, expense control, lease-up — rather than re-appraising.
- **Low-rate environments (roughly below 3.50%):** DSCR often produces more proceeds than LTV, making the LTV ceiling the binding constraint. In that environment, the loan is effectively a multiple of appraised value, and rate changes below a threshold barely move the needle on sizing.
- **Very high-rent markets:** The statutory limit can cut the loan when income and value would otherwise support a very large mortgage — Manhattan, San Francisco, and similar metros. In most markets it is not a factor.
- **Deals with cash-out potential:** When the cost of refinancing comes in below 80% LTV, the loan produces cash-out — and HUD caps that at 80% LTV rather than the standard 87%. Understanding where the cost-of-refinance criterion lands early tells you whether you're looking at a cash-out refinance or a transaction-cost-covering refinance, and which LTV ceiling actually applies.

There's no universal answer for which test will bind on your deal — it depends on the interplay of value, NOI, rate, unit mix, and the cost of getting the transaction done. The tool below is designed to give you a quick read on all four before you engage third-party reports.

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FREQUENTLY ASKED

Common Questions About *223(f) Loan Sizing*

How does HUD size a 223(f) loan?

HUD runs four independent tests and the loan is the lowest result across all four: an LTV test against appraised value, a DSCR test, the statutory per-unit mortgage limit, and the cost-of-refinance criterion. Market-rate deals cap at 87% LTV against a 1.15x minimum DSCR. No single test sets the number — whichever produces the smallest loan amount binds.

What is the maximum LTV on a HUD 223(f) loan?

87% of appraised value for market-rate acquisitions and refinances. On a refinance that produces cash-out, the ceiling drops to 80%. Fannie Mae and Freddie Mac are typically capped at 75% on comparable deals. LTV is only one of four tests, so a deal that clears 87% can still size lower on DSCR.

What DSCR does HUD require for a 223(f) loan?

1.15x is the industry shorthand. HUD sizes the DSCR-constrained loan by multiplying Debt Service NOI by 87% to arrive at maximum annual debt service, covering P&I and MIP — and 1 divided by 0.87 is approximately 1.149, so the two describe the same calculation. HUD also requires a 7% vacancy assumption on market-rate deals even where actual vacancy is lower.

What is the difference between Market NOI and Debt Service NOI in HUD underwriting?

HUD underwrites two NOIs on the same deal. Market NOI is what the property stabilizes to at market occupancy with normalized expenses; divided by the cap rate it produces appraised value, which drives the LTV test. Debt Service NOI — also called DSCR NOI — is the more conservative figure HUD derives from actual operating history and the appraiser's NOI, applying a 7% vacancy floor and the replacement reserve amount from the Property Capital Needs Assessment rather than the appraiser's $250 per unit. It feeds the DSCR test and is almost always lower.

Does the interest rate affect how much a HUD 223(f) loan sizes?

Only through the DSCR test — the LTV and statutory tests are rate-independent. HUD converts the debt service budget into a loan amount by dividing by an annual loan constant built from three components: the interest rate, the 0.25% MIP, and the amortization rate. At 5.00% on a 35-year term that constant is approximately 6.306%.

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HUD/FHA Multifamily Origination

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